

## MSCF Investments 46-972

Fall 2021 Mini 1

### Instructor:

Duane J. Seppi

### Contact Information:

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### Office Hours:

TBA (Zoom or phone), after class, or by appointment

### Weekly recitation sessions:

TBA

### Required Readings:

Lecture slides, course notes, and reproduced articles

### Suggested future reading (FYI only):

*Active Portfolio Management*, by Richard Grinold and Ronald Kahn  
*Investment Science*, by David Luenberger, Oxford University Press

### Learning Objectives:

This course is an introduction to the science & art of quantitative investment and the fundamental drivers of asset pricing. The main learning objective is to prepare you to manage portfolios using quant investment strategies and risk-management tools. This includes being able to:

- ❑ Construct widely used portfolios and trading strategies. These include mean-variance efficient, factor/index “mimicking”, factor “tilts,” zero-cost, long/short, and CPPI portfolios.
- ❑ Use regression analysis to measure investment risks and estimate expected returns.
- ❑ Compute and use performance measures such as Jensen’s alpha, style analysis, information ratio, Sharpe ratios and return-attribution decompositions based on strategic & dynamic asset class allocations and security selection.

We will also review foundational theoretical concepts and empirical evidence about asset pricing:

- ❑ Factor models of returns and systematic and idiosyncratic randomness including the Fama-French model and other extensions.
- ❑ The risk-reward trade-off and the CAPM and APT models of expected returns.

- ❑ Theories about financial information including the Efficient Market Hypothesis.
- ❑ Behavioral theories of investor sentiment and associated empirical measures.
- ❑ Empirical evidence on predictability, asset-pricing puzzles and anomalies.
- ❑ The organization of financial markets and empirical evidence on trade execution.

In addition, the class will review several current hot issues in asset management:

- ❑ COVID crisis
- ❑ Text data
- ❑ Exchange traded funds

### Homework:

Homework will be assigned weekly. You may do homework *individually* or in a *team* of two. If you work on a team, you should do all of the HWs together. Students working in teams should each submit their HW individually but be clear on Canvas and on the HW itself about who is on your team.

Late homework without prior permission will **not** be accepted. However, I will accommodate religious, job search, and other legitimate issues if notified in advance.

Do **not** submit spreadsheets or large amounts of raw statistical computer output as your homework. While portions of spreadsheets and computer output can be pasted (as text) into your answers, your answers should be written out in words and the output should be formatted for readability. Communication and presentation are an important part of the asset management business.

### Project:

One of the main points of focus in the course is on using empirical analysis as an input in the investment management process. In particular, this means using current financial information either to predict future returns and risks ex ante (prediction) or to account empirically for what drives returns ex post (attribution). Thus, teams of two (this should be your HW partner or someone else if you are doing the HWs alone) will do a project involving original empirical analysis. In terms of substance, this project will involve some basic data representation analysis (e.g., some sort of figure and a table of summary statistics) and some sort of regression analysis. The deliverable is a five page report. In your report, please be sure to **cite appropriately** any source material you based your analysis on. The report is **due** at the start of the final class in **week 7**.

### Exams:

There will be one 15-minute **quiz** at the start of class in **week 5** and a **final exam** but no midterm.

Since the quiz can only help your grade (by providing insurance against a bad final), **make-up quizzes are not possible** if you need to miss the quiz. Instructor permission

based on a valid reason (e.g., illness, travel for a job interview) is necessary if you have to miss the quiz.

The final will be closed-note, closed-book, closed-computer, and cumulative, but you may bring one 8 1/2" x 11" sheet of formulas with you.

### **Honor Code:**

You are encouraged to talk with each other about all aspects of the course — including about homework — but teams (and individual students working alone) should, in the end, write up and turn in their own work in their own words based on their own analysis (e.g., their own spreadsheets, programs). It is not permitted to consult previous HW solution keys or solutions from students in previous years.

If you are confused, come to my office hours. I don't want you to be confused. Let's work together so that you can learn this important material.

### **Grades:**

Course grades will be based on the statistic:  $\max\{(Q + F, 1.10 F) + P + H$  where  $Q$  and  $F$  are your quiz and final scores (out of 10 and 100 points respectively),  $P$  is your project score (out of 50 points) and  $H$  is your homework score (10 points per week).

### **Course Outline:**

#### □ Week 1:

##### Introduction

- Overview of investment management process

##### Return drivers and characteristics

- Cash flows and discount rates
- How COVID is affecting stock valuations in market
- Co-movement and linear factor models
- Example: 3- and 5-factor Fama-French

##### Regression primer

#### □ Week 2:

##### Return drivers and characteristics (continued)

- Volatility
- Campbell-Shiller decomposition

##### Expected returns

- Risk-reward trade-off
- CAPM

- Multi-factor risk and reward: Arbitrage Pricing Theory
- Economics of factor pricing
- Financial Economics
- Cross-section of expected returns
- Jensen's alpha
- Fama-MacBeth estimation

□ Week 3:

Information, sentiment, and errors

- Efficient Market Hypothesis
- Behavioral finance
- Sentiment
- Bubbles
- Empirical evidence: Event studies, anomalies, equity premium puzzle
- High frequency price discovery
- Possible sources of errors, mistakes, & lags

□ Week 4:

Portfolio construction

- Portfolio math
- Strategic, tactical and security selection decisions and return attribution
- Constant Proportion Portfolio Insurance (CPPI) strategies
- Mean-variance efficient portfolios
- Portfolio factor exposures
- Total return swaps and alpha transfer

□ Week 5:

**Quiz** on material from weeks 1-4

Trading

- Architecture of financial markets
- Limit order markets
- Components of bid/ask spreads
- High frequency trading

□ Week 6:

Performance evaluation

- Sharpe ratio
- Style analysis
- Cautions about manipulability of performance metrics

Current topics

- Text data

□ Week 7:

**Project report due**

Current topics

- ETFs

Active portfolio management

- Information ratios and Fundamental Law of Asset Management

Class wrap-up

□ **Final exam:** Friday, October 15<sup>th</sup> from 1:30 – 4:30 pm